```
  }}
}
```

❶ Sets five workers, as per the configuration

❷ Creates a router

❸ Creates a master with a router and a probe

❹ Repeats both expectations

NOTE The Akka test kit provides tools to create automated tests that form clusters in multiple JVMs, but developers rarely use them. The Akka Cluster codebase is full of them, as they are necessary to test the properties of the cluster in an automated way. However, this is a topic beyond the scope of this book. You can find more information at http://mng.bz/wvEa.

Dynamically growing and shrinking a simple application is simple with the `cluster` extension. Joining and leaving the cluster is easy, and you can test the functionality in a REPL console, a tool that allows you to experiment and verify how things work. If you've followed along with the REPL sessions, it should be apparent how solid this extension is; crashes in the cluster are properly detected.

Clustering is a notoriously painful chasm to cross, usually requiring many admin and programming changes. In this chapter, you've seen that Akka makes it much easier and doesn't require rewriting code. The word-counting example is about the generic way Akka processes jobs in a cluster in parallel. You used cluster routers and `ask` to make the actors cooperate and deal with failure, and you got a taste of how easy it would be to test the application. The ability to test the logic of a distributed application without any additional effort is a significant advantage that you quickly get used to with the Akka test kit: you can find problems in your application before you put it into operation on a large scale.